EE3032 Introduction to VLSI Design – Midterm

Fall 2007. Printed percentages sum to 110%, not 100%; all allocations are preserved. **1. (40%)** Answer the following problems.

(a) Draw a CMOS transistor-level schematic realizing the Boolean function $f = \overline{a(b+c)}$. (4%)

(b) List the process for transferring the mask pattern to the surface of a silicon region. (4%)

(c) List three approaches for increasing the performance of a CMOS circuit. (6%)

(d) List two latchup prevention approaches. (4%)

(e) List three principles for reducing the switching power of a CMOS circuit. (6%)

(f) Explain the purpose of design rules in VLSI design. (4%)

(g) Explain what the body effect is. (4%)

(h) Explain why adding a guard ring can reduce the latchup effect. (4%)

(i) Explain the energy consumption of an inverter when the inverter performs a switching. (4%)

2. (10%) (a) Design an 8-to-1 multiplexer using pass transistor logic. (5%) (b) Design a two-input exclusive-or (XOR) gate using CMOS transmission gates. (5%)

3. (20%) Complete the following problems.

(a) Figure 1(a) shows the XOR and OR gate. Assume that P_A and P_B are the mutually independent signal probabilities of inputs A and B to be at "1". Express the $P_{0\to1}$ of X and Y. (5%)

(b) Figure 1(b) shows a circuit with one OR and two AND gates. Assume the 1's probabilities of inputs all are 0.2. Calculate the $P_{0\to1}$ of A, B, and C. (10%)

(c) Figure 1(c) shows the pass transistor logics with NMOS and PMOS. Level the source and drain of the circuits, and calculate V_{out} of the circuits. (5%)

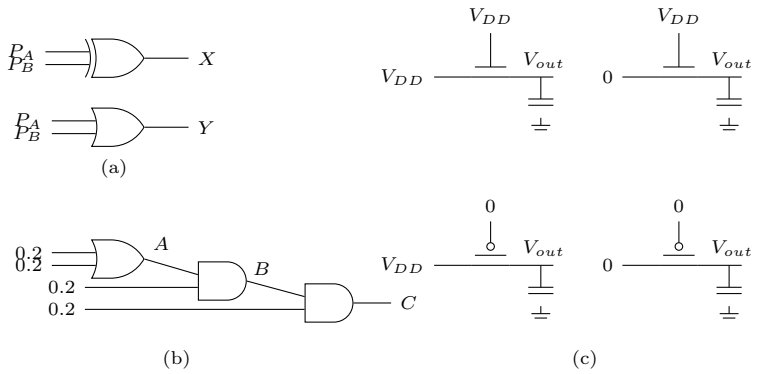


Figure 1: (a) primitive gates (b) cascade design (c) pass transistors.

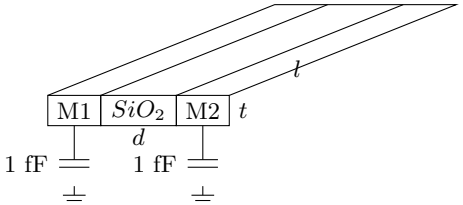


Figure 2: Two parallel wires.

4. (10%) There are two parallel wires, M1 and M2, as shown in Figure 2. Assume that $\epsilon_{SiO_2} = 5 \times 10^{-14}$ F/cm, $d = 1\,\mu$m, $t = 1\,\mu$m, and $l = 100\,\mu$m. If M2 is floating and M1 has a 0 V to 1.8 V transition, how much crosstalk noise will be coupled on M2?

5. (10%) Figure 3 depicts two designs of inverter chains. Assume that the equivalent resistance of a minimum size W/L NMOS is R and $\mu_n = 2\mu_p$. The gate capacitance of a W/L transistor is C_g. Also, the parasitic capacitances of drain/source and wires are neglected. Which inverter chain consumes lower short-circuit power? Explain your answer.

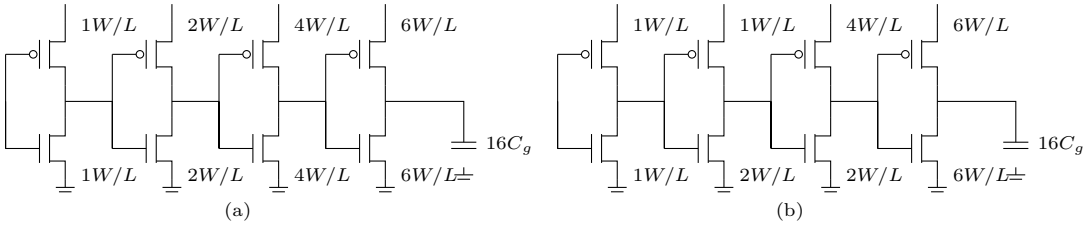
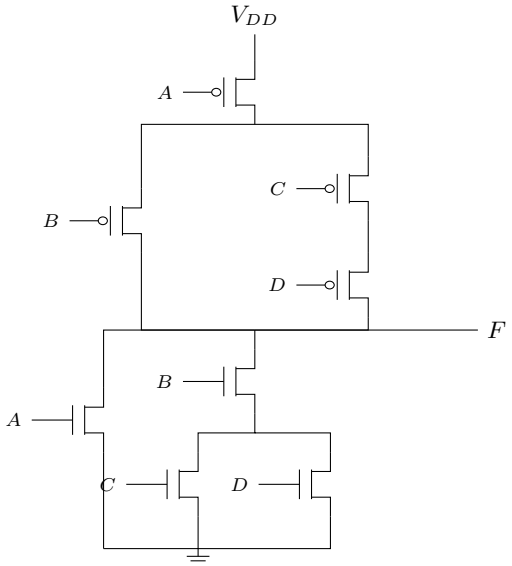


Figure 3: Inverter chains.

6. (10%) Suppose that the signal delay on an interconnect of length $50\,\mu$m is known to be 0.13 ps. If the line is increased to $100\,\mu$m, then what is the delay of the line? (4%) If you can insert an inverter in the $100\,\mu$m interconnect and the inverter delay is 0.2 ps, where should the inverter be inserted such that the wire delay is minimized, and how much is the delay of the interconnect with buffer insertion? (6%)

7. (10%) Size the transistors in the circuit shown below that implements a function $F = \overline{A + B(C + D)}$ so that its worst case current driving capability is equivalent to that of an inverter with $W_p/L_p = 10\lambda/2\lambda = 5$ and $W_n/L_n = 4\lambda/2\lambda = 2$. Assume that the channel length of all transistors is fixed at 2λ.



Editable reconstruction of all three source pages. Printed questions, values, connections and percentage labels are retained. The very small ps delays are reproduced as printed; no unit correction is assumed.